

W100/301

NATIONAL
QUALIFICATIONS
2002

FRIDAY, 18 JANUARY
9.00 AM – 10.10 AM

**MATHEMATICS
HIGHER**

Units 1, 2 and 3

Paper 1

(Non-calculator)

Read Carefully

- 1 Calculators may **NOT** be used in this paper.
- 2 Full credit will be given only where the solution contains appropriate working.
- 3 Answers obtained by readings from scale drawings will not receive any credit.

FORMULAE LIST

Circle:

The equation $x^2 + y^2 + 2gx + 2fy + c = 0$ represents a circle centre $(-g, -f)$ and radius $\sqrt{g^2 + f^2 - c}$.

The equation $(x - a)^2 + (y - b)^2 = r^2$ represents a circle centre (a, b) and radius r .

Scalar Product: $\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}| |\mathbf{b}| \cos \theta$, where θ is the angle between $\mathbf{a}$ and $\mathbf{b}$

$$\text{or } \mathbf{a} \cdot \mathbf{b} = a_1 b_1 + a_2 b_2 + a_3 b_3 \text{ where } \mathbf{a} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} \text{ and } \mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}.$$

Trigonometric formulae: $\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$

$$\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$$

$$\sin 2A = 2 \sin A \cos A$$

$$\cos 2A = \cos^2 A - \sin^2 A$$

$$= 2 \cos^2 A - 1$$

$$= 1 - 2 \sin^2 A$$

Table of standard derivatives:

$f(x)$	$f'(x)$
$\sin ax$	$a \cos ax$
$\cos ax$	$-a \sin ax$

Table of standard integrals:

$f(x)$	$\int f(x) dx$
$\sin ax$	$-\frac{1}{a} \cos ax + C$
$\cos ax$	$\frac{1}{a} \sin ax + C$

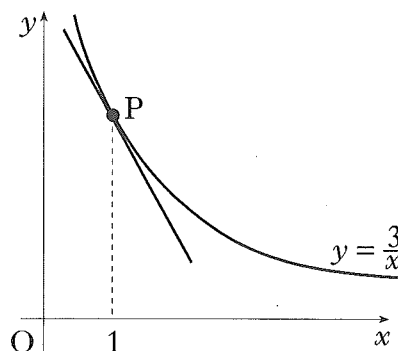
ALL questions should be attempted.

Marks

1. (a) Find the equation of the straight line through the points A(-1, 5) and B(3, 1). 2
- (b) Find the size of the angle which AB makes with the positive direction of the x -axis. 2

2. (a) If $u = \begin{pmatrix} 1 \\ 7 \\ -2 \end{pmatrix}$ and $v = \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}$, write down the components of $u + 3v$ and $u - 3v$. 2
- (b) Hence, or otherwise, show that $u + 3v$ and $u - 3v$ are perpendicular. 2

3. Find the equation of the tangent to the curve with equation $y = \frac{3}{x}$ at the point P where $x = 1$. 5

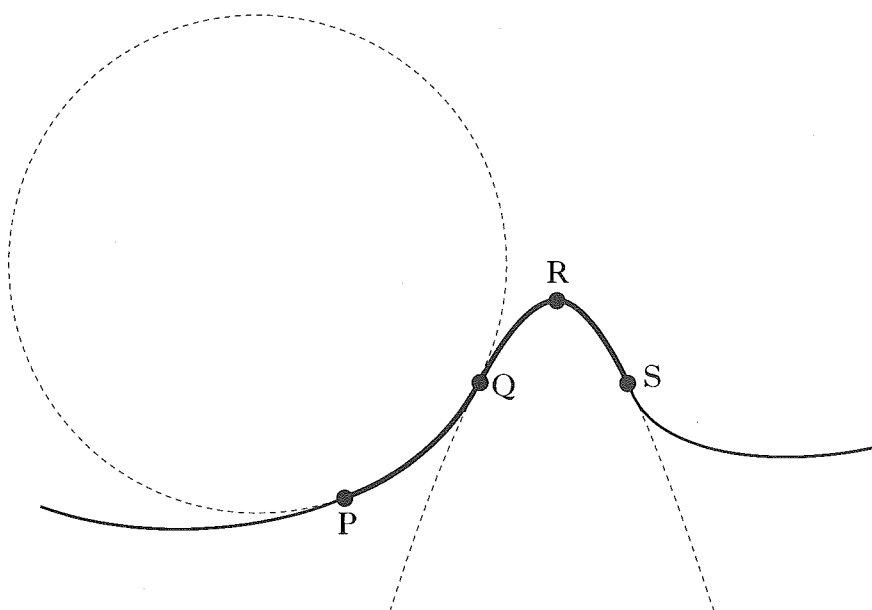


4. (a) Write down the exact values of $\sin\left(\frac{\pi}{3}\right)$ and $\cos\left(\frac{\pi}{3}\right)$. 1
- (b) If $\tan x = 4 \sin\left(\frac{\pi}{3}\right) \cos\left(\frac{\pi}{3}\right)$, find the exact values of x for $0 \leq x \leq 2\pi$. 2
5. Given that $(x - 2)$ and $(x + 3)$ are factors of $f(x)$ where $f(x) = 3x^3 + 2x^2 + cx + d$, find the values of c and d . 5

[Turn over]

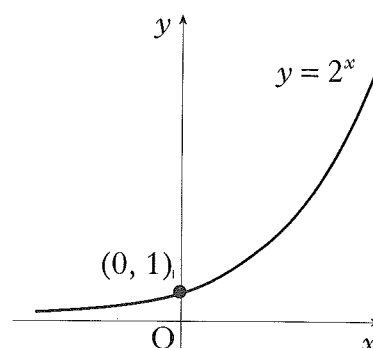
Marks

6. The side view of part of a roller coaster ride is shown by the path PQRS. The curve PQ is an arc of the circle with equation $x^2 + y^2 + 4x - 10y + 9 = 0$. The curve QRS is part of the parabola with equation $y = -x^2 + 6x - 5$. The point Q has coordinates (2, 3).



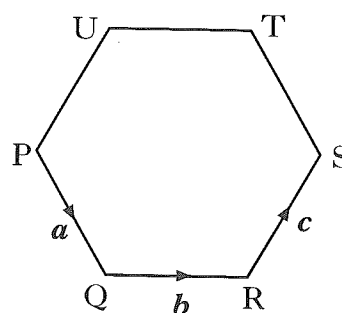
- (a) Find the equation of the tangent to the circle at Q. 4
- (b) Show that this tangent to the circle at Q is also the tangent to the parabola at Q. 2
7. Find $\int \left(\sqrt[3]{x} - \frac{1}{\sqrt{x}} \right) dx$. 4

8. The diagram shows part of the graph of $y = 2^x$.
- (a) Sketch the graph of $y = 2^{-x} - 8$. 2
- (b) Find the coordinates of the points where it crosses the x and y axes. 2



Marks

9. The function f , defined on a suitable domain, is given by $f(x) = \frac{3}{x+1}$.
- (a) Find an expression for $h(x)$ where $h(x) = f(f(x))$, giving your answer as a fraction in its simplest form. 3
- (b) Describe any restriction on the domain of h . 1
10. A function f is defined by $f(x) = 2x + 3 + \frac{18}{x-4}$, $x \neq 4$.
Find the values of x for which the function is increasing. 5
11. PQRSTU is a regular hexagon of side 2 units.
 $\vec{PQ}$, $\vec{QR}$ and $\vec{RS}$ represent vectors $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ respectively.
Find the value of $\mathbf{a} \cdot (\mathbf{b} + \mathbf{c})$. 3
12. If $\log_a p = \cos^2 x$ and $\log_a r = \sin^2 x$, show that $pr = a$. 3



[END OF QUESTION PAPER]